

Sheaves on Bun_G and the spectral action

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Lecture 1 F local field $\mapsto$ res. field $\mathbb{F}_q$ G/F conn. reductive. $k = \overline{\mathbb{F}_q}$

Perf = perfectoid spaces over $\overline{\mathbb{F}_q} = k$. Bun_G on Perf stack of G -bundles on X_{FF} .

$$\begin{array}{ccc} (D, \xi_1, \xi_2, d: \xi_1 \rightarrow \xi_2) & \mapsto & (D, \alpha|_{(D \times X_{\text{FF}})^\wedge}) \\ \swarrow h_1 & \nearrow \text{Hck}_G & \searrow \text{Hck}_G \\ \xi_1 & \text{Bun}_G & \text{Bun}_G \times \text{Div}^1 \\ & \searrow h_2 & \downarrow \\ & & (D, \xi_2) \end{array} \quad (R, R^+) \in \text{Perf}$$

Slogan - sheaves on Hck_G give FM kernels

- Source of sheaves on Hck_G is geometric Satake:
- Convolution diagram

$$\begin{array}{ccc} & \widetilde{\text{Hck}}_G & \\ \swarrow & & \searrow \\ \text{Hck}_G \times_{\text{Div}^1} \text{Hck}_G & & \text{Hck} \end{array}$$

Sheaf theory: Fix $\ell \neq p$ and Λ a $\mathbb{Z}_\ell[\sqrt{q}]$ -algebra

For X a small v -stack, we will use

$$\left[\begin{array}{l} D_{\text{rel}}(X) = D_{\text{rel}}(X, \Lambda) = D_{\text{mot}}^{\text{oc}}(X) \otimes_{D_{\text{mot}}^{\text{oc}}(\ast)} D(\Lambda) \\ D_{\text{et}}(X) = D_{\text{et}}(X, \Lambda) \text{ if } \ell \text{ is nilpotent in } \Lambda \end{array} \right.$$

$\uparrow$
see étale cohomology of diamonds

$D(\Lambda)$ = derived cat of Λ -modules

If ℓ nilp in Λ , $D_{\text{rel}}(X) \hookrightarrow D_{\text{ét}}(X)$ with explicit essential image.

Rem. $D_{\text{ét}}(*) \xrightarrow{\sim} D_{\text{ét}}(\text{Spd } C)$, $C = \widehat{F}$, very false for D_{rel} .

Main player: $D(\text{Bun}_G)$

Geometric Satake: Let $\widehat{G}/\Lambda$ be the dual group w/ its action of W_F ,

$$\widehat{G} \rtimes W_F = {}^L G.$$

Th. There is a subcat $\text{Sat}(G) \subset D(\text{Hck}_G)$ stable under convolution

+ an exact sym. monoidal equivalence $\text{Rep}(\widehat{G} \rtimes W_F) \xrightarrow{\sim} \text{Sat}(G)$

$\uparrow$

(1,1) cat. of reps on fin. proj. Λ modules

"i.e. $\widehat{G}$ acts algebraically and W_F continuously"

Rem $\text{Hck}_G \rightarrow \text{Div}^1$ and that is why W_F shows up.

In the étale setting, can base change along $\text{Spd } C \rightarrow \text{Div}^1$ to get

$$\text{Sat}_C(G) \xrightarrow{\sim} \text{Rep}(\widehat{G})$$

$$\cap \\ D(\text{Hck}_{G,C})$$

but not in D_{rel} .

So for $K \in \text{Sat}_G$, get a functor $T_K : D(\text{Bun}_G) \rightarrow D(\text{Bun}_G \times \text{Div}^1)$

$$F \mapsto h_{2*}(h_1^* F \otimes K)$$

In the étale setting, $D_{\text{ét}}(\text{Bun}_G) \xrightarrow{\sim} D(\text{Bun}_{G,C})$,

so get an endofunctor $T_K : D(\text{Bun}_G) \rightarrow D(\text{Bun}_{G,C}) \xleftarrow{\sim} D(\text{Bun}_G)$

Prop

$$\begin{array}{ccc}
 D_{\text{ét}}(\text{Bun}_G) & \xrightarrow{T_K} & D_{\text{ét}}(\text{Bun}_G \times \text{Div}^1) \xrightarrow{S^*} D_{\text{ét}}(\text{Bun}_G, c) \\
 & \searrow S_K & \uparrow S \\
 & & D_{\text{ét}}(\text{Bun}_G)
 \end{array}$$

$\text{Bun}_G \times \text{Spd } C$
 $\downarrow$
 $S: \text{Bun}_G \times \text{Div}^1$

- $S_K \circ S_{K'} = S_{K'} \circ S_K$

- $S_K \circ S_{K'} = S_{K \otimes K'}$ $K \otimes K'$ is monoidal str. on Sat .

$\Rightarrow S_K$ adjoint to $S_{K^\vee} \Rightarrow$ commutes with limits and colimits.

- S_K preserves LLA objects, preserves compact objects.

- S_K only depends on

$$\text{Sat}_G \simeq \text{Rep}(\hat{G} \rtimes W_F) \rightarrow |\text{Rep}(\hat{G})|$$

Spectral action: Recall Par_G stack of L-parameters for $\hat{G} \subseteq W_F$

$$\text{Par}_G \rightarrow \text{Spec } \Lambda / \hat{G}$$

Given $W \in \text{Rep}(\hat{G})$, get $\mathcal{W}$ v.b. on Par_G .

$\text{Perf}(\text{Par}_G)$ is the sym. monoidal cat. of perfect complexes on Par_G .

Th. There is an action of $\text{Perf}(\text{Par}_G)$ on $D_{\text{ét}}(\text{Bun}_G)$ s.t. for $W \in \text{Rep } \hat{G}$,

$$(P \mapsto S_P)$$

$$S_{W^\vee} = S_W$$

where S_W is the Hecke operator

assoc. to W

$$[q + \# \pi_0(\mathcal{Z}_G)]$$

How does this help?

$$\hat{\mathbb{Z}}^{\text{Spec}}(G, \Lambda)$$

Consequence. The ring $\text{End}_{\text{Par}_G}(\mathcal{O})$ acts compatibly on every object in $D_{\text{ét}}(\text{Bun}_G)$

Par. 3

For example, it acts on $\text{Det}(\text{Bun}_G^{[1]}) \subset \text{Det}(\text{Bun}_G)$

$$\uparrow$$

$$\text{Rep}_\Lambda^{\text{sm}}(G(F))$$

so gives a map $\mathcal{Z}^{\text{Spec}}(G, \Lambda) \rightarrow \mathcal{Z}(G(F), \Lambda)$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ \text{Spectral Bernstein} & & \text{Bernstein's} \\ \text{centre} & \uparrow & \text{mysterious centre} \end{array}$$

How is this action of $\text{Perf}(\text{Par}_G)$ constructed?

Take $w \in \text{Rep}(\hat{G} \rtimes W_F)$ and maps $1 \xrightarrow{\alpha} W, W \xrightarrow{\beta} 1$ in $\text{Rep}(\hat{G})$ $\gamma \in W_F$

Associated to this is an endomorphism of $\mathcal{O}$ on Par_G . $(W, \alpha, \beta, \gamma)$ is an "excursion datum"

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{\alpha} & W \\ & \downarrow \gamma & \\ W & \xrightarrow{\beta} & \mathcal{O} \end{array}$$

[Tautological $\phi: W_F \rightarrow L_G$ over Par_G ,
compose with $L_G \rightarrow GL(W)$ to let W_F
act on W]

Take $J \in \text{Det}(\text{Bun}_G)$, then $T_w J \in \text{Det}(\text{Bun}_G \times \text{Div}^1)$

$$\uparrow$$

$$\text{Det}(\text{Bun}_G)^{W_F}$$

$$\begin{array}{ccc} J & \xrightarrow{\alpha} & S_w(J) = T_w(J) \\ & \downarrow & \downarrow \gamma \\ & S_w(J) = T_w(J) & \xrightarrow{\beta} J \end{array}$$

excursion operator

In (V. Lafforgue, FS)

$\text{End}_{\text{Par}_G}(\mathcal{O})$ is gen'd by
excursion operators. $(\ell \neq \pi_0(\mathbb{Z}_G))$

Lecture 2. Charts and compact generation.

Charts are essential geometry (also show up in $\text{Bun}_G^{\text{mer}} \rightarrow \text{Bun}_G$)

We will define $\mathcal{M} \xrightarrow{\pi} \text{Bun}_G$ "moduli space of filtrations"

For $G = GL_n$, see Hansen and AWS group.

G -bundles on $X_S \leftrightarrow$ exact tensor functors $\text{Rep}_G \rightarrow \text{Vect}(X_S)$

$\text{Fil}_G \text{Vect}(X_S) = \{ (\mathcal{E}, \text{Fil}^\bullet \mathcal{E}) \mid \mathcal{E} \in \text{Vect}(X_S) \text{ and } \text{Fil}^\bullet \mathcal{E} \text{ is a collection}$

of subbundles $\mathcal{E}^{\leq \lambda} \subset \mathcal{E}$ indexed by $\mathbb{Q}$ s.t.

- $\lambda' \leq \lambda, \quad \mathcal{E}^{\leq \lambda'} \subset \mathcal{E}^{\leq \lambda} \quad (\text{increasing})$

- $\bigcup_{\lambda} \mathcal{E}^{\leq \lambda} = \mathcal{E} \quad (\text{exhaustive})$

- $\bigcap_{\lambda} \mathcal{E}^{\leq \lambda} = 0 \quad (\text{separated})$

This can be turned into an exact $\otimes$ -cat.

A $\mathbb{Q}$ -filtered G -bundle on X_S is an exact $\otimes$ -functor $\text{Rep}_G \rightarrow \text{Fil}_G \text{Vect}(X_S)$

Moduli stack of these is $\mathcal{M}_G^{\text{big}} \rightarrow \text{Bun}_G$.

Forgetful $\text{Fil}_G \text{Vect}(X_S) \rightarrow \text{Vect}(X_S)$

and an assoc. graded $\text{Fil}_G \text{Vect}(X_S) \xrightarrow{q} \text{Vect}(X_S)$

$$q(\mathcal{E}, \text{Fil}^\bullet \mathcal{E}) = \bigoplus_{\lambda} \frac{\mathcal{E}^{\leq \lambda}}{\mathcal{E}^{\leq \lambda^-}}, \quad \text{where } \mathcal{E}^{\leq \lambda^-} = \bigcup_{\lambda' < \lambda} \mathcal{E}^{\leq \lambda'}.$$

$\mathcal{M} \subset \mathcal{M}_G^{\text{big}}$ is the substack of exact $\otimes$ -functors where $\forall p: G \rightarrow GL_n, \quad \frac{p_* \mathcal{E}^{\leq \lambda}}{p_* \mathcal{E}^{\leq \lambda^-}}$ is semistable of slope $\lambda, \quad \forall \lambda$.

$\mathcal{M}$ comes with two functors

$$\mathcal{M} \xrightarrow{\pi} \text{Bun}_G$$

$$\begin{array}{ccc} i \nearrow & \downarrow q & \nearrow \text{obvious map} \\ & & \end{array}$$

$$\coprod_{g \in B(G)} \left[* / \underline{G(\mathcal{O}_D)} \right]$$

$$\sim \mathcal{M} = \coprod_b \mathcal{M}_b, \quad \pi_b, \quad q_b \quad \text{and} \quad \begin{array}{ccc} \tilde{\mathcal{M}}_b & \rightarrow & \mathcal{M}_b \\ \tilde{q}_b \downarrow & & \downarrow q_b \\ * & \rightarrow & [* / \underline{G_b(\mathcal{O}_D)}] \end{array}$$

Example. $G = GL_n$, $\xi_b = \xi_1 \oplus \xi_2$ where ξ_1, ξ_2 are semistable of slopes $\lambda_1 < \lambda_2$

$\tilde{\mathcal{M}}_b$ is the moduli space of ext's $0 \rightarrow \xi_1 \rightarrow \xi \rightarrow \xi_2 \rightarrow 0$

Equivalently, it represents $S \mapsto H^1(X_S, \xi_2^* \otimes \xi_1)$ so it is a negative

(absolute) BC space. $\leadsto q_b$ and $\tilde{q}_b$ are rep. in loc. spatial diamonds. and

cohomologically smooth.

$$n=2, \xi_1 = \mathcal{O}, \xi_2 = \mathcal{O}(1). \text{ then } \tilde{\mathcal{M}}_b = \left(\text{Spt } k((t)) \right)^\diamond$$

Prop The map q_b is rep. in loc. spat. diamonds, and coh smooth of rel. dim $\langle 2p - v_b \rangle$.

$\tilde{q}_b$ is an iterated fibration in negative BC spaces.



The map π_b is more difficult. In our example, $G = GL_n$, $\xi_b = \xi_1 \oplus \xi_2$, we have given $\xi \rightarrow \text{Bun}_G$ cor. to ξ , get $\text{Gr}(\xi) \rightarrow X_S$ and the fibre of π_b over ξ consists of certain sections of this Grassmannian. From this perspective, smoothness of

π_b seems difficult.

Th. The map π_b is rep. in loc. spat. diamonds, cohomologically smooth of $\dim \langle 2p, v_b \rangle$. Its image in $|\text{Bun}_G| \simeq B(G)^{\text{op}}$ is thus open.

Image consists of all $b' \in B(G)$ with $b' \leq b$.

Proof. Jacobian criterion of smoothness. (see FS).



Compact generation of $\text{Det}(\text{Bun}_G)$

Means: Generated by compact objects. (objects where $\text{Hom}(c, -)$ commutes with filtered colimits)

Who cares? c.g. $\Rightarrow$ dualizable in Pr.

Fact. $D(\text{Rep}_{\Lambda}^{\text{sm}} G(\mathcal{O}_p))$ is compactly generated by $c\text{-Ind}_K^{G(\mathcal{O}_p)} \Lambda$ for $K \subset G(\mathcal{O}_p)$ pro-p cpt open.

Fix $b, K \subset G(\mathcal{O}_p)$ cpt $\checkmark$ open, consider

$$h: \Lambda \rightarrow \left[\ast / \underline{\Lambda} \right] \rightarrow \left[\ast / \underline{G(\mathcal{O}_p)} \right]$$

$$\left[\tilde{M}_b / K \right] \xrightarrow{\text{étale}} M_b \xrightarrow{\pi_b} \text{Bun}_G$$

$\underbrace{\hspace{10em}}_{f_K}$

$$A_K^b := f_{K,!} f_K^! \Lambda$$

Th. The collection $\{A_K^b\}_{b,K}$ is a collection of cpt generators of $\text{Det}(\text{Bun}_G)$.

Sanity check. What is $i_b^* A_K^b \in D_{\text{ét}}(\text{Bun}_G^b) \simeq D(\text{Rep}_{\Lambda}^{\text{sm}} G(\mathcal{O}_p))$

Claim. This is $c\text{-Ind}_K^{G(\mathcal{O}_p)} \Lambda$.

proper base change.
so $i_b^* A_K^b = g_! g^! \Lambda$
 $\simeq c\text{-Ind}_K^{G(\mathcal{O}_p)} \Lambda$

$$\begin{array}{ccccc} \text{Proof. } \tilde{M}_b / K & \rightarrow & M_b & \xrightarrow{f_K} & \text{Bun}_G \\ \uparrow i_K & & \uparrow i & & \uparrow i^b \\ \left[\ast / K \right] & \rightarrow & \left[\ast / \underline{G(\mathcal{O}_p)} \right] & \xrightarrow{g} & \text{Bun}_G^b \end{array}$$

Proof of thm. Input is

Prop For $A \in \text{Det}(\tilde{M}_b / \kappa)$, then

$$R\Gamma(\tilde{M}_b / \kappa, A) \xrightarrow{\sim} R\Gamma(* / \kappa, i_{\kappa}^* A) = (i_{\kappa}^* A)^{\kappa}.$$

Slogan $\tilde{M}_b$ is henselian along $*$

Toy model. $(\text{Spec } k \llbracket t \rrbracket)^{\langle \rangle}$ is henselian along $*$

Pt. Take b, κ and $B \in \text{Det}(\text{Bun}_a)$

$$R\Gamma_{\text{hom}}(A_b^{\kappa}, B) \xrightarrow{\text{adj.}} R\Gamma_{\text{hom}}(t_{\kappa}^! \mathcal{L}, t_{\kappa}^! B)$$

$$\xrightarrow{\text{smoothness}} R\Gamma_{\text{hom}}(\mathcal{L}, t_{\kappa}^* B)$$

$$= R\Gamma(\tilde{M}_b / \kappa, t_{\kappa}^* B)$$

$$\xrightarrow{\text{henselian}} R\Gamma(* / \kappa, i_{\kappa}^* t_{\kappa}^* B)$$

$$= R\Gamma(* / \kappa, i_{b*} B)$$

$$= (i_{b*} B)^{\kappa}$$

$$(\)^{\kappa} = t_x \quad \text{where } t: * / \kappa \rightarrow * \quad \text{and } t_x = t_!$$

□